Cameron Craddock, PhD

Machine Learning Engineer

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PROFESSIONAL SUMMARY

Uber technical lead with experience building, optimizing, and deploying production AI systems, with skills spanning machine learning, embedded systems, cloud and distributed infrastructure, data science, and research.

SKILLS

Machine Learning & AI: PyTorch, machine learning (ML), production machine learning, neural network training and inference, inference optimization, CNNs, LSTMs, Transformers, quantization, pruning, model compression, Neural Architecture Search (NAS), supervised and unsupervised learning, statistical modeling, data science

Edge AI & Embedded Systems: embedded systems, neural network deployment, model deployment and optimization, hardware-aware optimization, power, latency, and memory optimization, ARM Ethos-U55/U85 neural processing units (NPUs), NXP HiFi4 digital signal processor (DSP), sensor fusion, digital signal processing

Performance & Instrumentation: high-resolution power measurement, system performance modeling, performance regression analysis, observability and performance monitoring, profiling, NPU profiling, logic analyzers, hardware instrumentation, automated performance testing, benchmarking, A/B experimentation

Software & Infrastructure: Python, C/C++, MATLAB, R, Bash, Pandas, Linux, cloud infrastructure, distributed systems, infrastructure engineering, high-performance computing (HPC), multi-GPU training, containers, continuous integration and continuous deployment (CI/CD), version control, automated testing, continuous testing, release engineering, open-source software development, AI-assisted software development, agentic engineering workflows

Domain Expertise: electromyography (EMG), wearable computing, biomedical signal processing, medical imaging (MRI, fMRI, PET)

PROFESSIONAL EXPERIENCE

Staff Machine Learning Engineer (IC6) | Meta Platforms Inc.

Remote | Aug 2022 - Present

Technical lead developing production machine learning systems for low-power wearable devices and neural interfaces. Work spans machine learning, embedded software, specialized AI processors, signal processing, hardware instrumentation, and system-level performance optimization.

- Lead a cross-organizational initiative focused on machine learning system performance, setting technical direction for measuring, predicting, and optimizing the power, latency, and memory requirements of always-on neural network inference. Team accomplishments include robust infrastructure for observability and monitoring of model power and performance during development and continuous integration (CI); methods for estimating model power and latency from high-level model representations to enable training-time performance feedback and neural architecture search; evaluation of new model features against system power constraints; investigation and root-cause analysis of performance regressions; and model optimization efforts to reduce power consumption.
- Architected and deployed high-resolution power-measurement and automated performance-testing infrastructure for low-power embedded systems, including custom instrumentation electronics, 200-kHz data acquisition, embedded-software integration, automated testing, and performance dashboards. Integrated power, latency, and memory measurements into the development process to detect performance regressions before release.
- Drove model and system-level optimizations to meet stringent latency and power constraints, including an inference architecture redesign that reduced neural network execution time from 25 ms to less than 8 ms; model and processor optimizations that reduced active-inference power by 42% and improved device battery life by 0.8 hours; and a redesign of the EMG and motion-sensor data-delivery pipeline that reduced system power by 20%. Developed a strategy for dynamically adjusting model execution frequency expected to reduce model power by up to 50%.
- Optimized and deployed neural networks on resource-constrained AI accelerators, resolving unsupported operations, model-lowering issues, and deployment blockers while improving model architecture, quantization, preprocessing, memory utilization, computational efficiency, and energy consumption on ARM Ethos-U55/U85 neural processing units (NPUs).
- Re-engineered the real-time EMG and motion-sensor processing pipeline to resolve a major system-performance issue blocking production, reducing synchronization error between sensor streams from approximately 110 ms to 4.5 ms while improving the efficiency and reliability of sensor fusion and signal processing.
- Developed an AI-native engineering workflow for machine learning and embedded-systems development, creating reusable services, agent skills, and automated workflows that connect AI agents with model-training, power-measurement, and performance-analysis infrastructure. Enabled AI agents to execute specialized workflows spanning model optimization, deployment, physical power measurement, and performance analysis.

- Led model productization and release engineering across multiple generations of wearable devices, developing on-device evaluation and data-quality systems, transitioning manual processes toward automated deployment and continuous testing, establishing performance feedback and release gates, and resolving release-blocking model and embedded-system issues.
- Onboarded and mentored junior engineers, providing technical guidance through architecture and code reviews, documentation, and hands-on support. Developed self-service tools and shared engineering practices that enabled engineers to work more independently and effectively across the machine learning and embedded-systems stack.

Research Engineering Manager | Indeed

Austin, Texas | Aug 2021 - Aug 2022

Hybrid technical lead and engineering manager leading a team developing recommender systems for job postings and Career Guide content while establishing a new data-science competency within the Career Guide organization.

- Led and supported six engineers developing recommendation and ranking approaches for job postings and Career Guide articles, providing technical direction while managing team execution and growth.
- Built data-science methods for Career Guide personalization and user-journey analysis, including article clustering, reader-profile modeling, and methods for tracking how users navigated and engaged with Career Guide content to improve recommendations for new and existing articles.

Associate Professor of Diagnostic Medicine (Computational Radiology) | The University of Texas at Austin, Dell Medical School

Austin, Texas | Sep 2017 - Jul 2021

Led federally supported artificial intelligence (AI) and computational neuroimaging research focused on diagnosis, prognosis, treatment prediction, open-source software development, and large-scale biomedical data systems.

- Led development of AI and neuroimaging methods, including fMRI-based brain-computer interfaces, neurofeedback systems, and large-scale longitudinal studies.
- Led open-source software development, including high-throughput neuroimaging analysis pipeline software and cloud and high-performance computing (HPC) neuroimaging pipelines for high-throughput processing and rapid clinical assessment.
- Led supporting software and data-collection efforts spanning mobile behavioral assessment, multimodal imaging, and MRI protocol optimization.

EDUCATION

PhD, Electrical & Computer Engineering | Georgia Institute of Technology | 2009

MS, Electrical & Computer Engineering | Georgia Institute of Technology | 2002

BS, Computer Engineering | Georgia Institute of Technology | 1999

OPEN SCIENCE LEADERSHIP

Brainhack - Co-founder of an open science organization that organizes hackathons to facilitate interdisciplinary collaborative approaches to neuroscience; over 150 events held since 2012.

Preprocessed Connectomes Project (PCP) - Founded the project to enable sharing of preprocessed neuroimaging data and accelerate reproducible analyses across the community.

Organization for Human Brain Mapping - Open Science SIG - Founding Chair, 2016

Organization for Human Brain Mapping - Education Chair - 2017-2019

GRANTS, AWARDS AND PUBLICATIONS

Author on over 98 peer-reviewed scientific publications and a patent (h-index: 58, i10-index: 98).

Invited speaker at over 35 conferences, workshops, and symposia including SXSW (2016 & 2017).

- 2021 - Organization for Human Brain Mapping Open Science Award ($2,500).
- 2014 - NIMH Biobehavioral Research Award for Innovative New Researchers ($2,600,000).
- 2010 - NARSAD Young Investigator Grant ($75,000).